

```
# Transformers =transformers is the Hugging Face Transformers library, which provides pre-trained AI models for
# Removes the existing version of the transformers library.

!pip uninstall -y transformers
# Installs a specific version (4.46.3) along with its required libraries.
!pip install -U transformers==4.46.3 \
    accelerate \
    sentencepiece \
    huggingface-hub==0.26.2
```

```
WARNING: Ignoring invalid distribution ~ransformers (/usr/local/lib/python3.12/dist-packages)
WARNING: Skipping transformers as it is not installed.
WARNING: Ignoring invalid distribution ~ransformers (/usr/local/lib/python3.12/dist-packages)
Collecting transformers==4.46.3
  Using cached transformers-4.46.3-py3-none-any.whl.metadata (44 kB)
Requirement already satisfied: accelerate in /usr/local/lib/python3.12/dist-packages (1.14.0)
Requirement already satisfied: sentencepiece in /usr/local/lib/python3.12/dist-packages (0.2.2)
Requirement already satisfied: huggingface-hub==0.26.2 in /usr/local/lib/python3.12/dist-packages (0.26.2)
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Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->acce
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Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests->transform
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests->tra
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests->tra
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2->torch>=2
Using cached transformers-4.46.3-py3-none-any.whl (10.0 MB)
WARNING: Ignoring invalid distribution ~ransformers (/usr/local/lib/python3.12/dist-packages)
Installing collected packages: transformers
WARNING: Ignoring invalid distribution ~ransformers (/usr/local/lib/python3.12/dist-packages)
Successfully installed transformers
```

```
# Import Model and Tokenizer from transformer Library
from transformers import AutoModelForCausalLM, AutoTokenizer
# Load model(Phi-3) and tokenizer
model = AutoModelForCausalLM.from_pretrained(
    "microsoft/Phi-3-mini-4k-instruct",
    # automatically choose the model CPU or GPU
    device_map="auto",
    # Choose the best data type automatically
    torch_dtype="auto",
    # Allow the model to run its own code
    trust_remote_code=
    True
    ,
    )
# Load the tokenizer for the model
tokenizer = AutoTokenizer.from_pretrained("microsoft/Phi-3-mini-4k-instruct")
```

WARNING:torchao.kernel.intmm:Warning: Detected no triton, on systems without Triton certain kernels will not work
 WARNING:transformers_modules.microsoft.Phi-3-mini-4k-instruct.f39ac1d28e925b323eae81227eaba4464caced4e.modeling_...
 WARNING:transformers_modules.microsoft.Phi-3-mini-4k-instruct.f39ac1d28e925b323eae81227eaba4464caced4e.modeling_...
 Loading checkpoint shards: 100% 2/2 [00:00<00:00, 2.33it/s]

```
#Prompt given to the model to perform the task
# <|assistant|> tells the model that the assistant need to start generating a response
prompt = "Write an email apologizing to Sarah for the tragic gardening mishap. Explain how it happened.<|assist
# Tokenizer convert input prompt into tokens(numbers) so that model can understand
input_ids = tokenizer(prompt,
# PyTorch is a Python library used to build and train machine learning and deep learning models.
# returns the tokens as pytorch tensors
# Input_ids =Extract the token IDs
#.to("cuda") moves the tokens to the GPU for fast processing
return_tensors="pt").input_ids.to("cpu")
# Generate the text
# Prints the token
for id in input_ids[0]:
    print(tokenizer.decode(id)) # Each token is converted back into text by using tokenizer.decode(id)
# Generate the response from the model
generation_output = model.generate(
input_ids=input_ids, # given input tokens to the model previously
max_new_tokens=20 # Genrated at most 20 new tokens
)
# Prints the response by converting token IDs back into normal text
print(tokenizer.decode(generation_output[0]))
# Decode the token ID 3323 to see which word it represents
print(tokenizer.decode(3323))
```

The attention mask is not set and cannot be inferred from input because pad token is same as eos token. As a con
 The `seen\_tokens` attribute is deprecated and will be removed in v4.41. Use the `cache\_position` model input ins
 `get\_max\_cache()` is deprecated for all Cache classes. Use `get\_max\_cache\_shape()` instead. Calling `get_max_cac
 WARNING:transformers_modules.microsoft.Phi-3-mini-4k-instruct.f39ac1d28e925b323eae81227eaba4464caced4e.modeling_
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 <|assistant|>
 Write an email apologizing to Sarah for the tragic gardening mishap. Explain how it happened.<|assistant|> Subje

 Dear
 Sub

```
#List of background colors
colors_list = [
'102;194;165', '252;141;98', '141;160;203',
'231;138;195', '166;216;84', '255;217;47'
```

```

]
# Function to display how a tokenizer divide sentence into tokens
def show_tokens(sentence, tokenizer_name):
    # Load the tokenizer from Hugging face model
    tokenizer = AutoTokenizer.from_pretrained(tokenizer_name)
    # Convert the sentence into token IDs
    token_ids = tokenizer(sentence).input_ids
    # loop for tokens
    # idx = Position of each token
    # t= token IDs
    for idx, t in enumerate(token_ids):
        # Print the each token with different color
        # decodes the tokens to the normal text
        print(f'\x1b[0;30;48;2;{colors_list[idx % len(colors_list)]}m' # idx % len(colors_list) cycles through the
              + tokenizer.decode(t) +
              '\x1b[0m', # \x1b[0m resets all text formatting (color, background color, bold, etc.) back to normal.
              end=' '
              )

```

```

from transformers import AutoModel, AutoTokenizer
# Load a tokenizer
tokenizer = AutoTokenizer.from_pretrained("microsoft/deberta-base")
# Load a language mode named DEBERTA
# The model processes token IDs and produces the embeddings(Numerical representations)
model = AutoModel.from_pretrained("microsoft/deberta-v3-xsmall")
# Tokenize the sentence into token IDs and return them as pytorch tensors
tokens = tokenizer('Hello world', return_tensors='pt')
# **tokens unpacks the dictionary into input_ids, attention_mask, etc.
# [0] extracts the last hidden state (the embedding of each token).
'''model(
    input_ids=tokens["input_ids"],
    attention_mask=tokens["attention_mask"]
)
'''
output = model(**tokens)[0]
# Display the shape(dimensions)
output.shape

```

```
torch.Size([1, 4, 384])
```

```

# tokens['input_ids'][0] used to select the first sentence from the batch
# if we use tokens['input_ids'] we will select tensor([[1, 7592, 2088, 2]]) there we have two brackets by using
for token in tokens['input_ids'][0]:
    # Decodes the token IDs into normal text
    print(tokenizer.decode(token))

```

```

[CLS]
Hello
world
[SEP]

```

TEXT EMBEDDING

```

# installs Sentence-transformers library
!pip install sentence-transformers

```

```

WARNING: Ignoring invalid distribution ~ransformers (/usr/local/lib/python3.12/dist-packages)
Requirement already satisfied: sentence-transformers in /usr/local/lib/python3.12/dist-packages (5.6.0)
Collecting transformers<6.0.0, >=4.41.0 (from sentence-transformers)
  Using cached transformers-5.13.1-py3-none-any.whl.metadata (32 kB)
Requirement already satisfied: huggingface-hub>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers)
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Requirement already satisfied: click<9.0.0,>=8.4.2 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub)
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Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformers)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformers)
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Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2->torch>=1.11.0->sentence-transformers)
Requirement already satisfied: shellingsham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->transformers)
Requirement already satisfied: rich>=13.8.0 in /usr/local/lib/python3.12/dist-packages (from typer->transformers)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->transformers)
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Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=13.8.0)
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Using cached transformers-5.13.1-py3-none-any.whl (11.5 MB)
WARNING: Ignoring invalid distribution ~ransformers (/usr/local/lib/python3.12/dist-packages)
Installing collected packages: transformers
WARNING: Ignoring invalid distribution ~ransformers (/usr/local/lib/python3.12/dist-packages)
Successfully installed transformers

```

```

# is a Linux shell command that deletes the Hugging Face cache folder
!rm -rf ~/.cache/huggingface

```

```

# Imports SentenceTransformer class from sentence_transformers library
# A SentenceTransformer converts and entire sentence or paragraph into numerical vectors(Embeddings)
from sentence_transformers import SentenceTransformer
# Load model to generate meaningful embeddings
model = SentenceTransformer("sentence-transformers/all-mpnet-base-v2")
# Convert text to text embeddings
# The model understand the meaning of sentence and represent it as numerical vectors
vector = model.encode("Best movie ever!")
# Display the shape(dimensions)
vector.shape

```

```

modules.json: 100%                               349/349 [00:00<00:00, 30.5kB/s]

config_sentence_transformers.json: 100%             116/116 [00:00<00:00, 6.55kB/s]

README.md:      11.6k/? [00:00<00:00, 842kB/s]

sentence_bert_config.json: 100%                    53.0/53.0 [00:00<00:00, 4.86kB/s]

config.json: 100%                                571/571 [00:00<00:00, 53.4kB/s]

model.safetensors: 100%                          438M/438M [00:05<00:00, 41.4MB/s]

tokenizer_config.json: 100%                       363/363 [00:00<00:00, 30.2kB/s]

vocab.txt:      232k/? [00:00<00:00, 6.66MB/s]

tokenizer.json:  466k/? [00:00<00:00, 22.0MB/s]

special_tokens_map.json: 100%                     239/239 [00:00<00:00, 20.2kB/s]

config.json: 100%                                190/190 [00:00<00:00, 17.9kB/s]

(768,)

```

```

# Gensim is a python library for Natural language processing (NLP)
!pip install gensim

```

```

WARNING: Ignoring invalid distribution ~ransformers (/usr/local/lib/python3.12/dist-packages)
Collecting gensim
  Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (8.4 kB)
Requirement already satisfied: numpy>=1.18.5 in /usr/local/lib/python3.12/dist-packages (from gensim) (2.0.2)
Requirement already satisfied: scipy>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from gensim) (1.16.3)
Requirement already satisfied: smart_open>=1.8.1 in /usr/local/lib/python3.12/dist-packages (from gensim) (7.6.1)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart_open>=1.8.1->gensim)
Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (27.9 MB)
    27.9/27.9 MB 24.7 MB/s eta 0:00:00
WARNING: Ignoring invalid distribution ~ransformers (/usr/local/lib/python3.12/dist-packages)
Installing collected packages: gensim
WARNING: Ignoring invalid distribution ~ransformers (/usr/local/lib/python3.12/dist-packages)
Successfully installed gensim-4.4.0

```

```

# Import the downloader module from the Gensim library.
# It is used to download pre-trained word embedding models.
import gensim.downloader as api
# Download embeddings (66MB, glove, trained on wikipedia, vector size: 50)
# "glove-wiki-gigaword-50" means:
# - GloVe = Global Vectors for Word Representation
# - Trained on Wikipedia and Gigaword text
# Other options include "word2vec-google-news-300"
# More options at https://github.com/RaRe-Technologies/gensim
# Each word is represented by a vector of 50 numbers
model = api.load("glove-wiki-gigaword-50")
# topn=11 means finds the 11 words which are similar to king
#model['king'] return the Embedding vector of word king
#most_similar compare the word king with all the words and returns
model.most_similar([model['king']], topn=11)

```

```

[=====] 100.0% 66.0/66.0MB downloaded
[('king', 1.0000001192092896),
 ('prince', 0.8236179351806641),
 ('queen', 0.7839043140411377),
 ('ii', 0.7746230363845825),
 ('emperor', 0.7736247777938843),
 ('son', 0.766719400882721),
 ('uncle', 0.7627150416374207),
 ('kingdom', 0.7542161345481873),
 ('throne', 0.7539914846420288),
 ('brother', 0.7492411136627197),
 ('ruler', 0.7434253692626953)]

```

```

#Imports pandas library
# Pandas used to store and analyze the data
import pandas as pd
# Import request from urllib library
# It is used to download data from the URL
from urllib import request
# Get the playlist dataset file
#Downloads the playlist dataset from ther internet
# request.urlopen open the URL and return the file
data = request.urlopen('https://storage.googleapis.com/maps-premium/dataset/yes_complete/train.txt')
#Diplay the information about the downloaded file
print(data.info)
# Parse the playlist dataset file. Skip the first two lines as
# decode("utf-8") Converts the downloaded bytes into readdable text
# split('\n') Seperate the text into individual lines
# [2:] skips the first two lines because they contain metadata, not playlist data
lines = data.read().decode("utf-8").split('\n')[2:]
# Remove playlists with only one song
#rstrip() remove the extra space at the end of the line
#split() Seperate the song IDs based on space
#keep only the playlist that contain more than 1 song
playlists = [s.rstrip().split() for s in lines if len(s.split())
> 1]
# Load song metadata file
# A metadata file stores descriptive information about the main data.
songs_file = request.urlopen('https://storage.googleapis.com/maps-premium/dataset/yes_complete/song_hash.txt')
# read the metadata file and convert it into text
songs_file = songs_file.read().decode("utf-8").split('\n')
# Split the each line using tab character ('\t')
songs = [s.rstrip().split('\t') for s in songs_file]
# Create a Pandas DataFrame from the song metadata.
# The DataFrame will have three columns: id ,title,artist
songs_df = pd.DataFrame(data=songs, columns = ['id', 'title', 'artist'])
# set the id coloumn as index ,which may used for fast accessing
songs_df = songs_df.set_index('id')

```

<bound method HTTPResponse.info of <http.client.HTTPResponse object at 0x7d82fdaacd60>>

```

# print the first playlist and second playlist
print( 'Playlist #1:\n ', playlists[0], '\n')
print( 'Playlist #2:\n ', playlists[1])

```

```

Playlist #1:
['0', '1', '2', '3', '4', '5', '6', '7', '8', '9', '10', '11', '12', '13', '14', '15', '16', '17', '18', '19',

Playlist #2:
['78', '79', '80', '3', '62', '81', '14', '82', '48', '83', '84', '17', '85', '86', '87', '88', '74', '89', '9

```

```

#Import word2vec model from gensim.models library
from gensim.models import Word2Vec
# Train our Word2Vec model
model = Word2Vec(
playlists,# Input data (previous playlist )
vector_size=32, # Each song is represented by a vector of size 32 number s
window=20,# Look at upto 20 neighbouring songs in the library
negative=50, # Uses 50 negative sample during training
min_count=1, # Include every song even it appears 1 time
workers=4 # uses 4 CPU cores to train the model faster
)
# select the song Id to which we want recommend
song_id = 2172
# Ask the model for songs similar to song #2172
# positive=str(song_id) tells the model to search using this song by converting interger song Id to String beca
model.wv.most_similar(positive=str(song_id))

```

```

[('2886', 0.9985059499740601),
 ('3167', 0.9982885718345642),

```

```
( '2849', 0.9980040788650513),
( '2976', 0.9966935515403748),
( '2640', 0.9964846968650818),
( '3094', 0.9961196184158325),
( '2715', 0.9959449768066406),
( '3136', 0.9958121180534363),
( '2881', 0.9956096410751343),
( '11517', 0.9955428838729858)]
```

```
# Selects the song at row index 2172 from the songs_df DataFrame using iloc and prints all the details of that
print(songs_df.iloc[2172])
```

```
title      Fade To Black
artist      Metallica
Name: 2172 , dtype: object
```

```
#Import numpy library
# numpy is used to easu to calculate arrays
import numpy as np
#
def print_recommendations(song_id):
    # Finds the 5 songs that are similar to the given song
    # most_similar() returns a list of tuples:
    # (song_id, similarity_score)
    similar_songs = np.array(model.wv.most_similar(
        positive=str(song_id)# Convert song Id into string
        ,topn=5)
   )[: ,0] # Extract only the song IDs
    # Use the recommended song IDs to retrieve the song title and artist
    # from the songs DataFrame.
    return songs_df.iloc[similar_songs]
# Display recommendations by calling the function
print_recommendations(2172)
```

	title	artist	
id			
2886	The Zoo	Scorpions	
3167	Unchained	Van Halen	
2849	Run To The Hills	Iron Maiden	
2976	I Don't Know	Ozzy Osbourne	